

```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

```
In [2]: df=pd.read_csv("CrimesOnWomenData.csv")
df.head()
```

```
Out[2]:
```

	Unnamed: 0	State	Year	Rape	K&A	DD	AoW	AoM	DV	WT
0	0	ANDHRA PRADESH	2001	871	765	420	3544	2271	5791	7
1	1	ARUNACHAL PRADESH	2001	33	55	0	78	3	11	0
2	2	ASSAM	2001	817	1070	59	850	4	1248	0
3	3	BIHAR	2001	888	518	859	562	21	1558	83
4	4	CHHATTISGARH	2001	959	171	70	1763	161	840	0

Data Cleaning

```
In [3]: #Removing Unnecessary column
df.drop(columns='Unnamed: 0',inplace=True)
```

```
In [4]: #Rename columns
df.rename(columns={
    'K&A':'Kidnapping_Abduction',
    'DD':'Dowry_Deaths',
    'AoW':'Assault_on_Women',
    'AoM':'Assault_on_Minors',
    'DV':'Domestic_Violence',
    'WT':'Women_Trafficking'
}, inplace=True)
```

```
In [5]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 736 entries, 0 to 735
Data columns (total 9 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   State                                736 non-null    str
1   Year                                736 non-null    int64
2   Rape                                736 non-null    int64
3   Kidnapping_Abduction                736 non-null    int64
4   Dowry_Deaths                        736 non-null    int64
5   Assault_on_Women                    736 non-null    int64
6   Assault_on_Minors                   736 non-null    int64
7   Domestic_Violence                   736 non-null    int64
8   Women_Trafficking                   736 non-null    int64
dtypes: int64(8), str(1)
memory usage: 51.9 KB
```

```
In [6]: df.isnull().sum()
```

```
Out[6]: State          0
Year          0
Rape          0
Kidnapping_Abduction  0
Dowry_Deaths    0
Assault_on_Women  0
Assault_on_Minors  0
Domestic_Violence  0
Women_Trafficking  0
dtype: int64
```

```
In [7]: #The statistical summary helps understand the distribution and scale of crime ca
df.describe().T
```

```
Out[7]:
```

	count	mean	std	min	25%	50%	75%
Year	736.0	2011.149457	6.053453	2001.0	2006.00	2011.0	2016.0
Rape	736.0	727.855978	977.024945	0.0	35.00	348.5	1069.0
Kidnapping_Abduction	736.0	1134.542120	1993.536828	0.0	24.75	290.0	1216.0
Dowry_Deaths	736.0	215.692935	424.927334	0.0	1.00	29.0	259.0
Assault_on_Women	736.0	1579.115489	2463.962518	0.0	34.00	387.5	2122.2
Assault_on_Minors	736.0	332.722826	806.024551	0.0	3.00	31.0	277.5
Domestic_Violence	736.0	2595.078804	4042.004953	0.0	13.00	678.5	3545.0
Women_Trafficking	736.0	28.744565	79.999660	0.0	0.00	0.0	15.0



```
In [8]: df.duplicated().sum()
```

```
Out[8]: np.int64(0)
```

```
In [9]: df['State'].nunique()
```

```
Out[9]: 70
```

```
In [10]: df['State']=df['State'].str.title()
df['State'].unique()
```

```
Out[10]: <StringArray>
[   'Andhra Pradesh', 'Arunachal Pradesh',      'Assam',
      'Bihar',      'Chhattisgarh',      'Goa',
      'Gujarat',      'Haryana', 'Himachal Pradesh',
'Jammu & Kashmir',      'Jharkhand',      'Karnataka',
      'Kerala',      'Madhya Pradesh',      'Maharashtra',
      'Manipur',      'Meghalaya',      'Mizoram',
      'Nagaland',      'Odisha',      'Punjab',
      'Rajasthan',      'Sikkim',      'Tamil Nadu',
      'Tripura',      'Uttar Pradesh',      'Uttarakhand',
      'West Bengal',      'A & N Islands',      'Chandigarh',
      'D & N Haveli',      'Daman & Diu',      'Lakshadweep',
      'Puducherry',      'Telangana',      'D&N Haveli',
      'Delhi Ut']
Length: 37, dtype: str
```

```
In [11]: df['State'] = df['State'].replace({
      'Delhi UT': 'Delhi',
      'D & N Haveli': 'Dadra and Nagar Haveli',
      'D&N Haveli': 'Dadra and Nagar Haveli',
      'A & N Islands': 'Andaman and Nicobar Islands',
      'Andaman And Nicobar': 'Andaman And Nicobar Islands'
})
```

```
In [12]: df['State'].unique()
```

```
Out[12]: <StringArray>
[   'Andhra Pradesh',      'Arunachal Pradesh',
      'Assam',      'Bihar',
      'Chhattisgarh',      'Goa',
      'Gujarat',      'Haryana',
'Himachal Pradesh',      'Jammu & Kashmir',
      'Jharkhand',      'Karnataka',
      'Kerala',      'Madhya Pradesh',
      'Maharashtra',      'Manipur',
      'Meghalaya',      'Mizoram',
      'Nagaland',      'Odisha',
      'Punjab',      'Rajasthan',
      'Sikkim',      'Tamil Nadu',
      'Tripura',      'Uttar Pradesh',
      'Uttarakhand',      'West Bengal',
'Andaman and Nicobar Islands',      'Chandigarh',
      'Dadra and Nagar Haveli',      'Daman & Diu',
      'Lakshadweep',      'Puducherry',
      'Telangana',      'Delhi Ut']
Length: 36, dtype: str
```

Feature Engineering

```
In [13]: df['Total_cases'] = df[['Rape', 'Kidnapping_Abduction', 'Dowry_Deaths',
      'Assault_on_Women', 'Assault_on_Minors', 'Domestic_Violence',
      'Women_Trafficking']].sum(axis=1)
```

Data Analysis

ResearchQeus 1.Which states report the highest and lowest total crimes?

```
In [14]: states_total_crime=df.groupby('State').agg({'Total_cases':'sum'})
highest_crime_state=states_total_crime.sort_values("Total_cases",ascending=False)
lowest_crime_state=states_total_crime.sort_values("Total_cases").head(10)
```

```
In [15]: highest_crime_state
```

```
Out[15]:
```

	Total_cases
--	-------------

State	
Uttar Pradesh	529734
Madhya Pradesh	413157
West Bengal	409242
Andhra Pradesh	391123
Rajasthan	379264
Maharashtra	365632
Assam	291980
Kerala	202986
Odisha	194774
Bihar	189886

```
In [16]: lowest_crime_state
```

```
Out[16]:
```

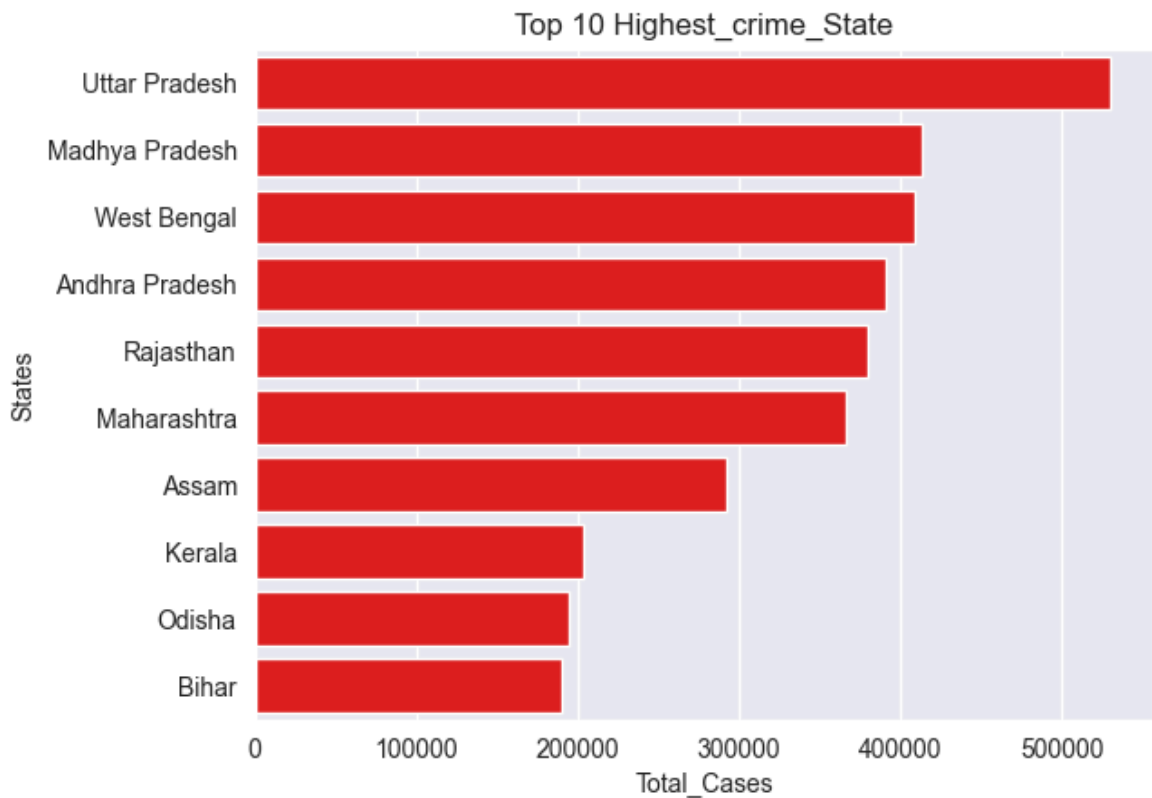
	Total_cases
--	-------------

State	
Lakshadweep	72
Puducherry	1793
Andaman and Nicobar Islands	1868
Mizoram	2844
Goa	4179
Meghalaya	4294
Manipur	4390
Arunachal Pradesh	4822
Chandigarh	5341
Daman & Diu	7015

```
In [17]: sns.barplot(data=highest_crime_state,x='Total_cases',
                    y='State',color='red')
```

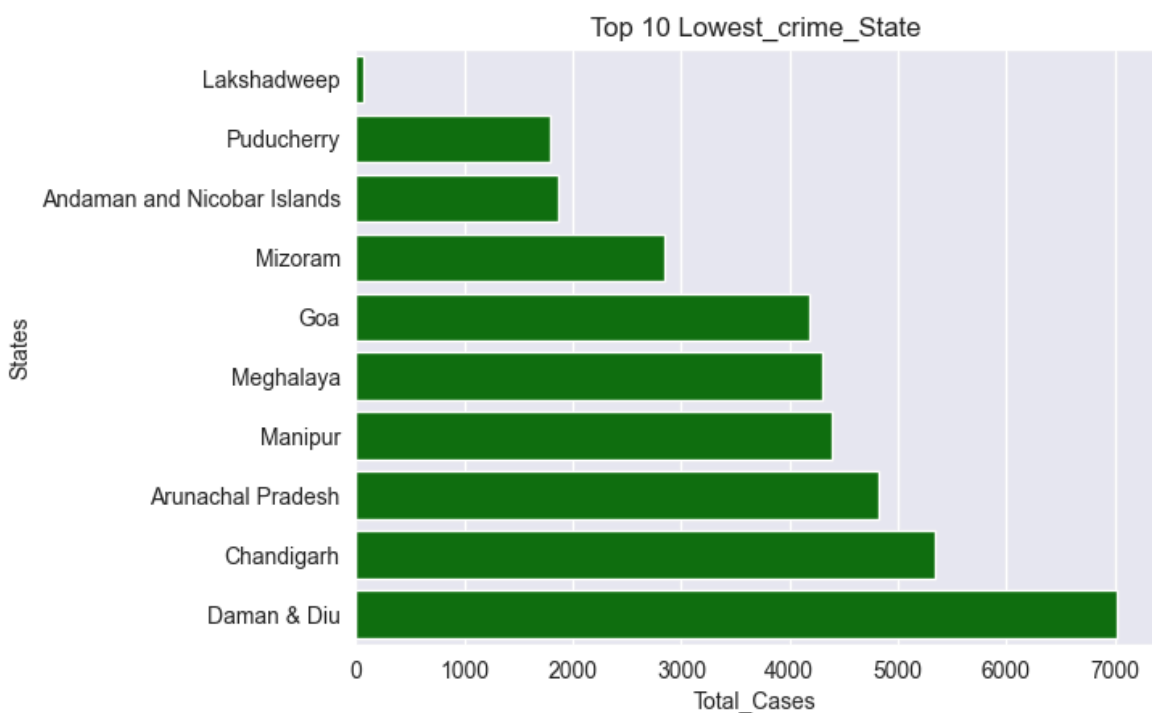
```
plt.xlabel('Total_Cases')
plt.ylabel('States')
plt.title('Top 10 Highest_crime_State')

plt.show()
```



```
In [18]: sns.barplot(data=lowest_crime_state,x='Total_cases',y='State',color='green')
plt.xlabel('Total_Cases')
plt.ylabel('States')
plt.title('Top 10 Lowest_crime_State')

plt.show()
```



Insights

- Uttar Pradesh reported the highest total crime cases, followed by Madhya Pradesh and West Bengal.
- Lakshadweep recorded the lowest number of total crime cases among all states and union territories.
- Large and densely populated states such as Uttar Pradesh, Madhya Pradesh, and Maharashtra showed significantly higher reported crime counts.
- Several northeastern states and smaller union territories reported comparatively lower crime cases.

ResearchQues 2: Which year recorded the highest crime activity?

```
In [19]: year_grouped=df.groupby('Year').agg({'Total_cases':'sum'})
year_grouped.head()
```

```
Out[19]:
```

	Total_cases
Year	

Year	
2001	128537
2002	128972
2003	128142
2004	140072
2005	139333

```
In [20]: plt.figure(figsize=(14,6))

sns.lineplot(
    data=year_grouped,
    x='Year',
    y='Total_cases',
    marker='o',
    linewidth=3,
    color='crimson'
)

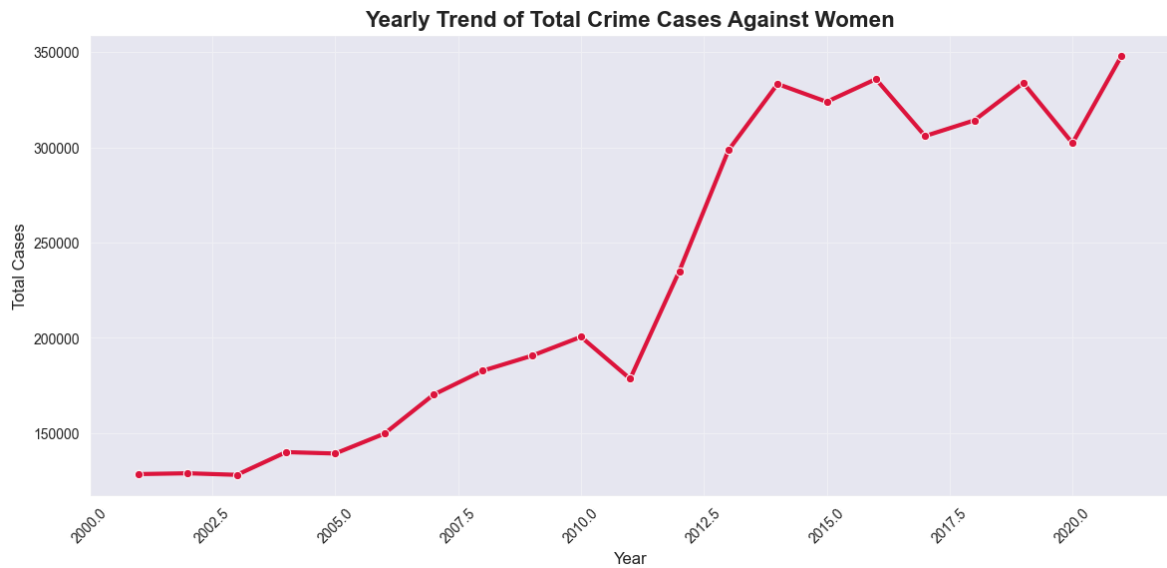
plt.title('Yearly Trend of Total Crime Cases Against Women',
          fontsize=16,
          fontweight='bold')

plt.xlabel('Year', fontsize=12)
plt.ylabel('Total Cases', fontsize=12)

plt.xticks(rotation=45)
```

```
plt.grid(alpha=0.3)

plt.show()
```



Insights

- From 2000 to 2010 , total crime cases against women showed a steady upward trend across India.
- A slight declined was observed between 2010 and 2011.
- Crime cases increased sharply from 2012 to 2015 , reaching one of the highest levels during entire period.

ResearchQues 3: Which crime category contributes most to total crimes?

```
In [21]: crime_column=df.iloc[:,2:9]
```

```
In [36]: crime_total=crime_column.sum()
crime_total=crime_total.sort_values(ascending=False).reset_index()
crime_total.columns=["Category", "TotalCases"]
crime_total
```

```
Out[36]:
```

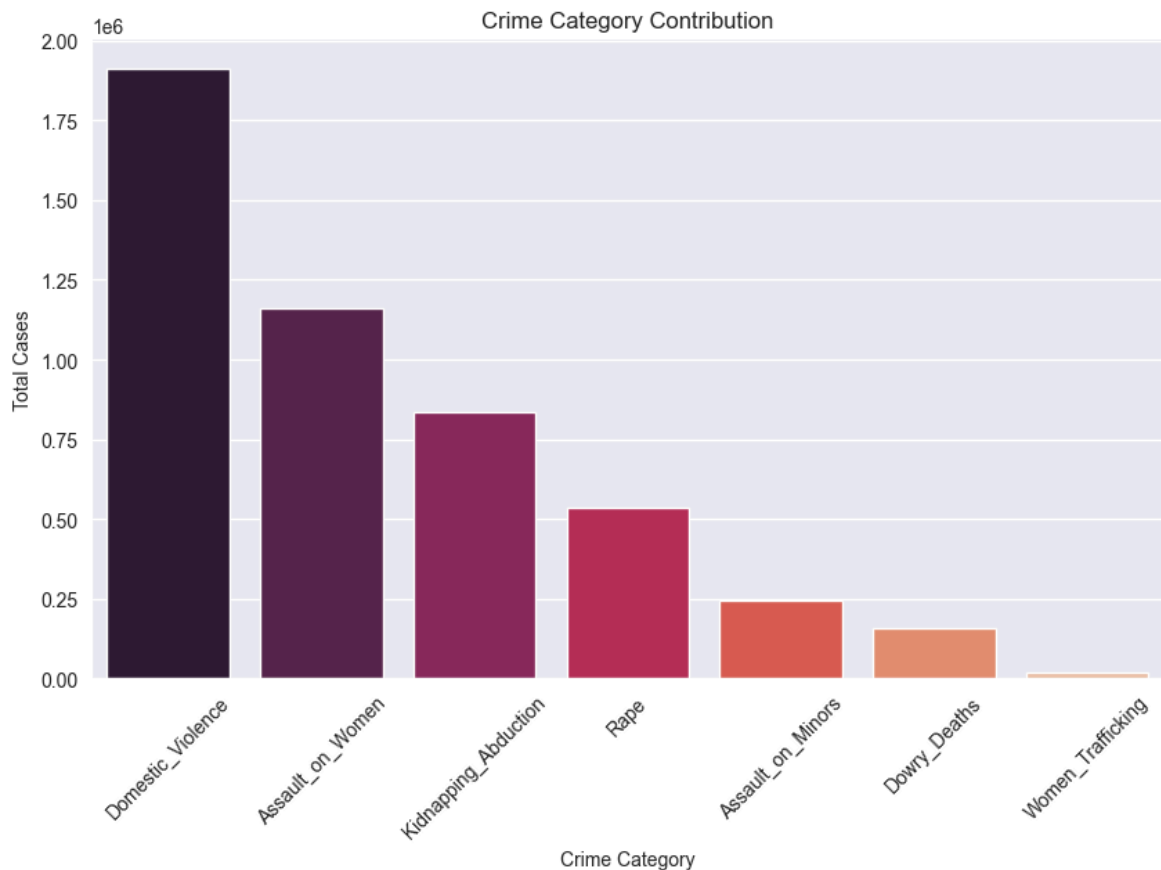
	Category	TotalCases
0	Domestic_Violence	1909978
1	Assault_on_Women	1162229
2	Kidnapping_Abduction	835023
3	Rape	535702
4	Assault_on_Minors	244884
5	Dowry_Deaths	158750
6	Women_Trafficking	21156

```
In [23]: plt.figure(figsize=(10,6))

ax=sns.barplot(crime_total,x='Category',y='TotalCases',palette='rocket',hue='Cat

plt.title('Crime Category Contribution')
plt.xlabel('Crime Category')
plt.ylabel('Total Cases')
plt.xticks(rotation=45)

plt.show()
```



Insights

- Domestic Violence contributed the highest share among all crime categories, accounting for ~40% of total reported crime cases.
- Women Trafficking recorded the lowest number of reported cases among all categories.
- The large contribution of domestic violence indicates that household-related violence remains a major concern across India.

ResearchQues 4: How has assault on minors changed over time?

```
In [24]: minors_assault=df.groupby('Year').agg(Total_case=('Assault_on_Minors','sum')).re
minors_assault
```

Out[24]:

	Year	Total_case
0	2001	9656
1	2002	10027
2	2003	12220
3	2004	9871
4	2005	9759
5	2006	9822
6	2007	10783
7	2008	12084
8	2009	10891
9	2010	9881
10	2011	8570
11	2012	9173
12	2013	12589
13	2014	21938
14	2015	24041
15	2016	27344
16	2017	7451
17	2018	6992
18	2019	6939
19	2020	7065
20	2021	7788

```
In [25]: plt.figure(figsize=(14,6))

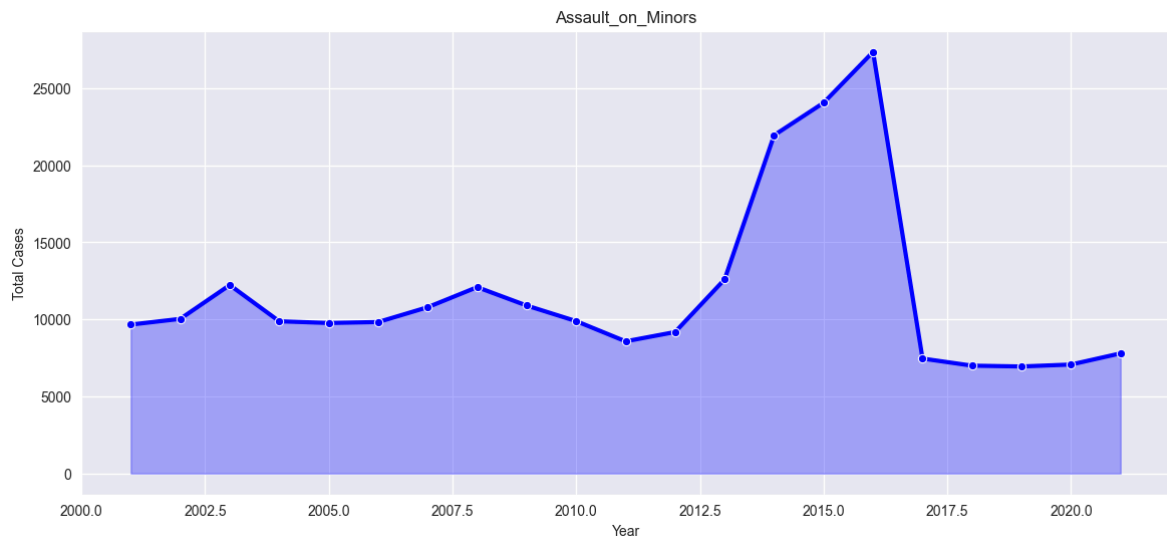
sns.lineplot(data=minors_assault,
              x='Year',
              y='Total_case',
              marker='o',
              linewidth=3,
              color='blue')

plt.fill_between(
    minors_assault['Year'],
    minors_assault['Total_case'],
    alpha=0.3,
    color='blue'
)

plt.title('Assault_on_Minors')
plt.xlabel('Year')
plt.ylabel('Total Cases')
```

```
# plt.xticks(rotation=45)
```

```
plt.show()
```



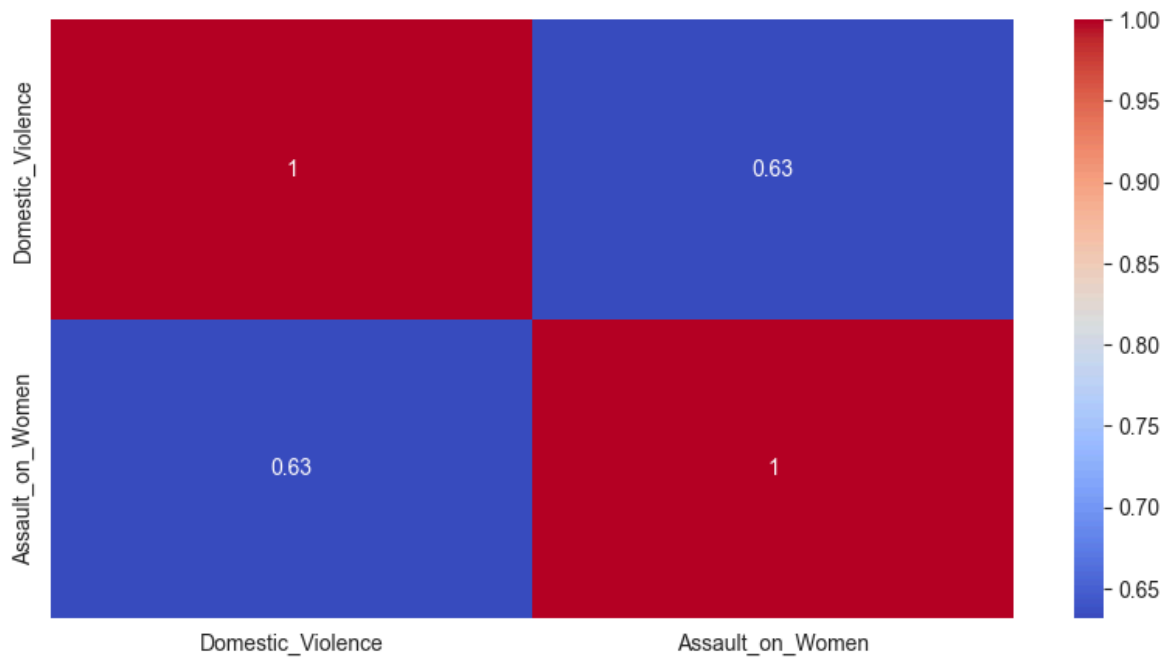
Insights

- Assault on minors remained relatively stable between 2001 and 2012.
- A sharp increase was observed from 2013 to 2016, with cases reaching peak levels during this period.
- After 2016, reported cases dropped significantly and remained comparatively lower in subsequent years.
- The trend suggests that assault on minors experienced sudden fluctuations rather than a steady long-term increase.

ResearchQues 5: Is domestic violence associated with assault on women?

```
In [26]: relation=df[['Domestic_Violence','Assault_on_Women']].corr()
```

```
In [27]: plt.figure(figsize=(10,5))  
  
sns.heatmap(data=relation,annot=True,cmap='coolwarm')  
plt.show()
```



Insights

- Domestic Violence and Assault on Women showed a moderate positive correlation of 0.63.
- This indicates that states reporting higher domestic violence cases also tend to report higher assault on women cases.
- The relationship suggests a possible association between the two crime categories, although correlation does not confirm direct causation.

ResearchQues 6 :Do a few states contribute to most crime cases?

```
In [40]: state_crime=df.groupby('State').agg(Total_case=('Total_cases','sum')).reset_index()
state_crime.head()
```

```
Out[40]:
```

	State	Total_case
0	Andaman and Nicobar Islands	1868
1	Andhra Pradesh	391123
2	Arunachal Pradesh	4822
3	Assam	291980
4	Bihar	189886

```
In [41]: overall_total=state_crime['Total_case'].sum()
overall_total
```

```
Out[41]: np.int64(4867722)
```

```
In [42]: state_crime['contribution%']=(state_crime['Total_case']/overall_total)*100
```

```
In [43]: top10_states=state_crime.sort_values('contribution%',ascending=False).head(10)
top10_states
```

```
Out[43]:
```

	State	Total_case	contribution%
33	Uttar Pradesh	529734	10.882585
19	Madhya Pradesh	413157	8.487687
35	West Bengal	409242	8.407259
1	Andhra Pradesh	391123	8.035032
28	Rajasthan	379264	7.791406
20	Maharashtra	365632	7.511357
3	Assam	291980	5.998288
17	Kerala	202986	4.170041
25	Odisha	194774	4.001338
4	Bihar	189886	3.900921

```
In [44]: top10_states_contribution=round(top10_states['contribution%'].sum(),2)
top10_states_contribution
```

```
Out[44]: np.float64(69.19)
```

```
In [45]: others_contribution=100-top10_states_contribution
others_contribution
```

```
Out[45]: np.float64(30.810000000000002)
```

```
In [39]: # DonutChart
labels = list(top10_states['State']) + ['Others']

sizes = list(top10_states['contribution%']) + [others_contribution]

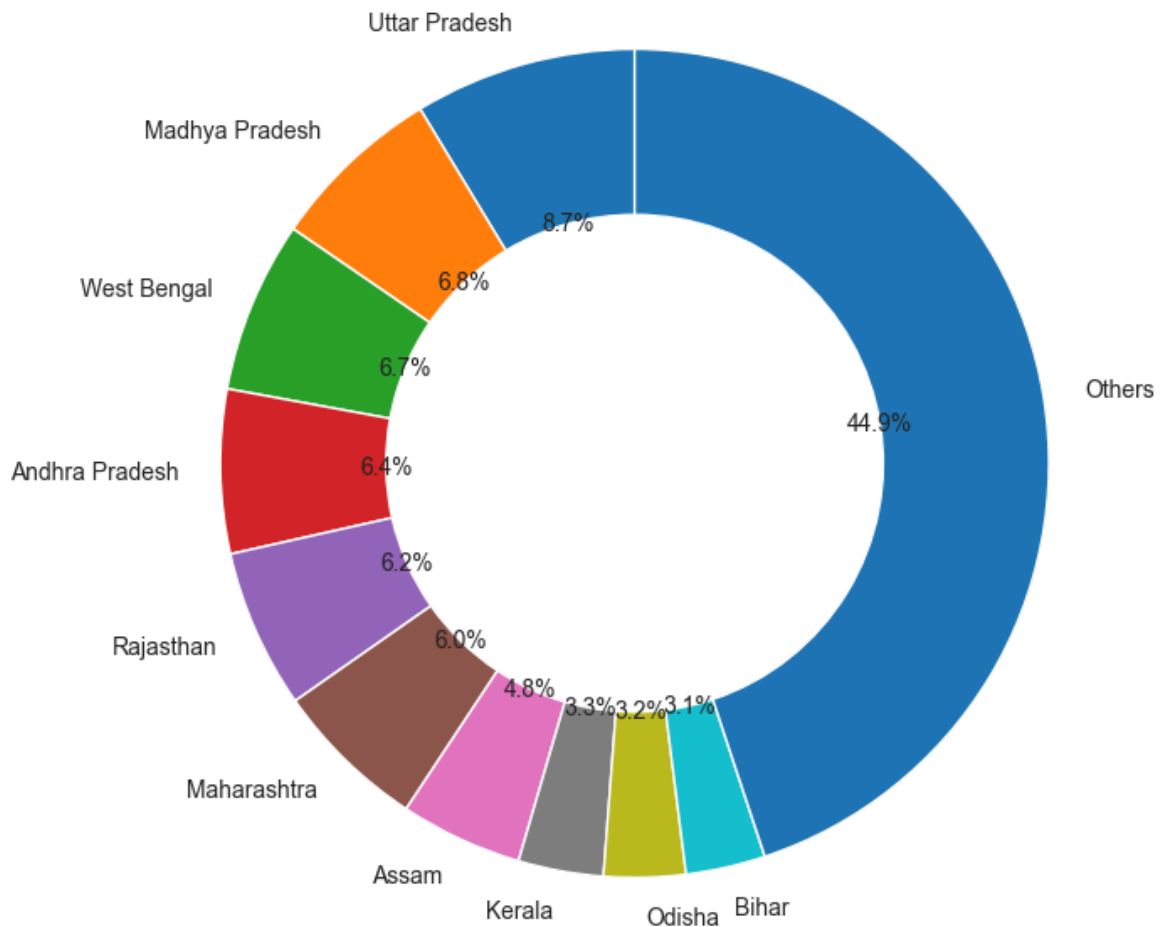
plt.figure(figsize=(10,8))

plt.pie(
    sizes,
    labels=labels,
    autopct='%1.1f%%',
    startangle=90,
    wedgeprops={'width':0.4}
)

plt.title('Crime Contribution by Top 10 States')

plt.show()
```

Crime Contribution by Top 10 States



Insights

- Uttar Pradesh contributed the highest share of total crime cases (~10.9%).
- The top 10 states together contributed nearly 69% of the total crime cases.
- This indicates that a significant portion of crimes is concentrated within a limited number of states.

Conclusion

- Uttar Pradesh recorded the highest overall crime cases among all states.
- Domestic violence was the largest contributing crime category.
- Crime cases generally increased over the years, especially after 2012.
- Assault on women and domestic violence showed a moderate positive relationship (correlation ≈ 0.63).
- A significant share of total crime cases was concentrated within the top 10 states.

- Some northeastern and union territory regions reported comparatively lower crime counts.

Future Scope

- Population-normalized crime rate analysis can provide more accurate comparisons.
- Predictive modeling can be applied to forecast future crime trends.
- Geographic visualization using maps can improve regional analysis.

In []: